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An Adaptive Inhibitory WSe₂ Transistor for Retinomorphic In-Sensor Image Processing

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ABSTRACT

Retinomorphic vision systems have emerged as a promising approach for low-power in-sensor image processing by mimicking the human retina. Retinal contrast enhancement amplifies local visual features at the sensor level, eliminating the need for computationally intensive external processing. However, most conventional retinomorphic devices require illumination-dependent bias adjustments, which introduce decision-making overhead and limit scalable array integration. Here, we report an adaptive inhibitory WSe₂ transistor that performs decision-free, intensity-adaptive image processing within a single pixel. The current inhibition ratio decreases monotonically from ~16.2 to ~1.2 as the back-gate voltage increases from 3 to 13 V, reflecting regime-dependent carrier transport. When integrated with photodiodes, the transistor switches between subthreshold and accumulation regimes depending on local illumination. Under a common top-gate pulse, dim pixels are strongly suppressed while bright pixels largely retain their current. We then processed gradient and noisy “G” patterns on the 5 × 5 photodiode array, achieving contrast enhancement and noise filtering. Furthermore, simulations of license-plate recognition under rain and fog show that processing restores character morphology, improving classification accuracy from 47.7% to 97.3%. These results show that adaptive inhibition offers a scalable, energy-efficient route to adaptive in-sensor processing for edge-vision hardware in low-visibility environments.

1 | Introduction

Object recognition under uncontrolled illumination remains a central challenge in machine vision systems deployed in real-world applications such as autonomous driving, robotics, and drones [1–6]. These conditions typically arise from shadows,

light scattering, backlighting, and low-light environments, where reduced contrast and increased background noise jointly degrade recognition accuracy [6–11]. Therefore, image processing such as contrast enhancement and background suppression is essential to ensure reliable object recognition [11–13]. In conventional vision systems, however, this processing is performed at external

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processors, requiring image sensors to first capture and transmit the optical data through a physically separated architecture [14–17]. Since state-of-the-art vision sensors generate raw data at rates of tens of Gb/s, transmitting this volume to the external processor often results in excessive energy consumption, latency, and system bottlenecks, particularly in resource-constrained edge devices [15, 17–20]. To address these limitations, in-sensor image processing has recently emerged as an alternative that performs sensing and computation at the sensor level, reducing data movement and improving energy efficiency [21–24].

Among various in-sensor architectures, retinomorphic vision systems, which mimic the visual processing of the human retina, have emerged as one of the most promising approaches for low-level visual feature enhancement [25–33]. This promise originates from the retina's ability to extract and enhance critical visual features with compact and energy-efficient processing, which conventional digital processors cannot match [25, 34–36]. This ability is rooted in “lateral inhibition”, in which inhibitory interactions between neighboring cells enhance the contrast between adjacent visual signals (Figure S1 and Note S1) [29, 37–40]. In this mechanism, stronger illumination at one location induces stronger inhibitory signals in neighboring cells, while weaker illumination produces proportionally weaker inhibition. This proportional inhibition produces a center-surround response, in which the contrast ratio at the input is amplified at the output, sharpening edge definition before downstream processing [39]. Crucially, this enhancement arises intrinsically from the retina's architecture without explicit computation, making retinomorphic devices particularly attractive for compact, low-power edge vision [25, 41, 42].

Such intrinsic processing has motivated diverse hardware implementations of lateral inhibition, ranging from multi-transistor circuits to single-device architectures. Yildirim et al. demonstrated a multi-transistor approach, realizing lateral-inhibition contrast enhancement with six MOS transistors per pixel [38]. Liao et al. used a single MoS₂ phototransistor with a compact pixel design, achieving 199 dB dynamic range and bias-tunable contrast enhancement [5]. Similarly, Kwon et al. implemented selectable excitatory and inhibitory modes in an adjustable synaptic phototransistor by modulating the gate bias [43]. However, in such approaches the gate bias must be deliberately adjusted to each illumination condition, making contrast enhancement a decision-dependent process. This requirement for external decision-making demands dedicated control circuitry, which adds latency and complicates large-scale array integration [4, 44]. For scalable array integration, it would instead be advantageous to obtain contrast enhancement without such a decision step, with the degree of inhibition set by the local illumination within each pixel. To the best of our knowledge, such decision-free, illumination-adaptive inhibition has not been demonstrated within a single pixel. Achieving such autonomous inhibition requires a pixel whose intrinsic response encodes the mapping between input illumination and inhibitory output, a constraint unmet by existing single-pixel implementations.

Here, we report an adaptive inhibitory WSe₂ transistor that performs decision-free, adaptive in-sensor image processing within a single pixel. We term this operating principle “adaptive inhibition”, in which each pixel autonomously inhibits its

own response according to local illumination, without coupling to neighboring pixels. This in-pixel operation reduces the hardware overhead associated with inter-pixel signal routing, interconnections, and computation. Image-processing functions can therefore be distributed by complexity, with simple contrast adjustment performed at the sensor and spatial operations such as edge or contour extraction left to the downstream processor. In this design, a photodiode is cascaded with a four-terminal WSe₂ transistor (Figure S2), where the photo-induced back-gate bias shifts the transistor between subthreshold and accumulation regimes, producing intensity-adaptive inhibition without external control. We then extend this operation to a 5 × 5 integrated array, where contrast enhancement and background suppression are validated using a gradient illumination pattern and a “G” character pattern, respectively. Building on this validated principle, we further simulate the application of adaptive-inhibition-based processing to license plate images captured under low-visibility conditions such as rain and fog, where it markedly improves character distinguishability. These processed images are evaluated by an artificial neural network (ANN) simulation, where the recognition accuracy increases from approximately 47.7% for the original rain/fog-degraded images to 97.3% after processing. These findings show that the proposed adaptive inhibitory system can serve as a practical in-sensor processing platform for adaptive vision hardware.

2 | Results and Discussion

2.1 | Adaptive Inhibitory Retinomorphic In-Sensor Image Processor Mimicking the Human Retina

The human retina is known to enhance contrast intrinsically through a three-layer cellular architecture comprising photoreceptors, horizontal cells, and bipolar cells (Figure 1a) [38, 45]. Within this pathway, incident light is transduced by photoreceptors, laterally suppressed by horizontal cells that link adjacent photoreceptors in proportion to neighboring activity, and subsequently relayed downstream by bipolar cells. This inter-cellular feedback produces a center-surround response that amplifies local intensity contrast while preserving global illumination context. Replicating this intrinsic, energy-efficient processing in conventional hardware is challenging because sensing and inhibition must remain tightly coupled at the pixel level [38–40]. In this study, we implement a hardware analogue of this retinal organization by cascading sensing and inhibitory elements at the pixel level, thereby enabling decision-free contrast enhancement directly at the image plane. As illustrated in the block diagram in Figure 1b, decision-required processing involves sequential steps such as condition decision and parameter selection. In contrast, the proposed decision-free scheme performs adaptive processing using fixed parameters across varying illumination conditions. Here, the pulse parameters are determined once from the device characteristics and are not reselected during operation. Such decision circuitry can introduce hardware overhead, including increased power consumption and footprint, as further discussed in Figure S3 and Note S2.

To realize this retinal function in hardware, we cascade a photodiode array with a four-terminal WSe₂ transistor array at the pixel

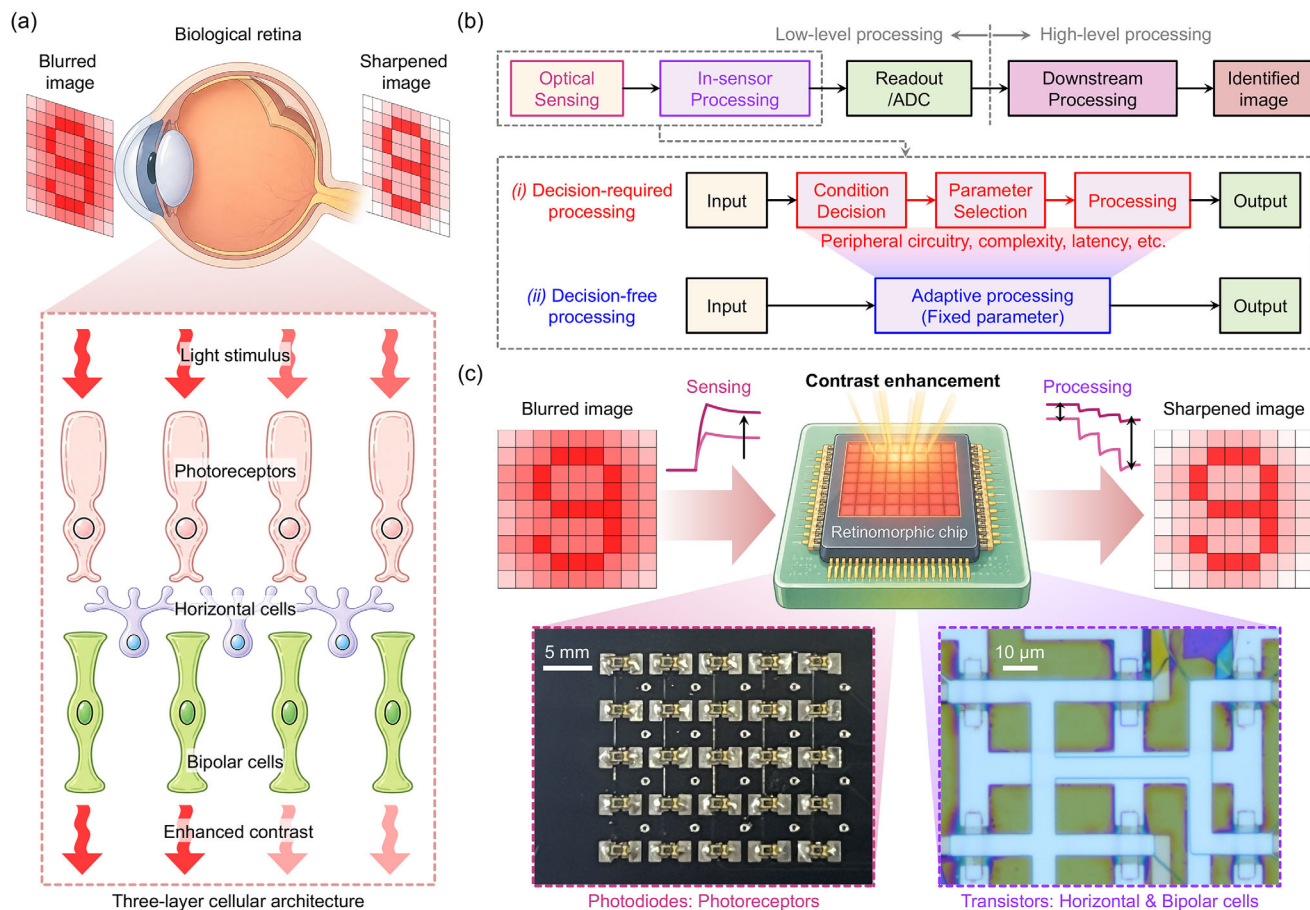


FIGURE 1 | Retinomorph in-sensor image processor mimicking the human retina. (a) Schematic of the three-layer cellular architecture of the biological retina (photoreceptors, horizontal cells, and bipolar cells). (b) Conceptual block diagrams illustrating the overall image-processing flow from optical sensing to image identification, highlighting low-level processing within the sensor hardware. The lower diagrams compare decision-required and decision-free processing architectures. (c) Schematic of the proposed retinomorph in-sensor image processor cascading a photodiode array with a four-terminal WSe₂ adaptive inhibitory transistor array. The bottom panels show the photodiode array on a PCB (left) and the fabricated transistor array (right).

level, as shown in Figure 1c. In this architecture, the photodiode replaces the photoreceptor for optical-to-electrical conversion, while the cascaded adaptive inhibitory transistor provides the inhibitory stage that follows it. The inhibition is generated within each pixel from its own illumination level, and therefore corresponds to the inhibitory function of the retinal pathway rather than to the lateral, neighbor-coupled interaction of horizontal cells. WSe₂ is chosen for its broad ambipolar operating range [46, 47], with the four-terminal configuration enabling independent biasing of the channel by the back and top gates. This separates sensing and processing into two stages, so that the photodetection front-end and the inhibitory stage can be optimized independently. Although the WSe₂ channel is itself photosensitive (Figure S4), its photoresponse alone provides only a limited variation in inhibition across illumination conditions (Figure S5), because the inhibition strength is governed by the back-gate-controlled transport regime. Cascading the photodiode therefore allows the incident illumination to set the transport regime directly, so that illumination-adaptive inhibition is obtained without adjusting the gate bias for each condition.

A blurred input image is first sensed by the photodiode array, which converts the incident light at each pixel into an electrical voltage that is applied to the back gate (BG) of the corresponding WSe₂ transistor as a steady bias. This illumination-dependent BG voltage determines the transport regime of the transistor. Under weak illumination, the low BG voltage places the transistor in the subthreshold regime, where the channel exhibits high sensitivity to a single transient pulse applied to the top gate (TG), resulting in strong inhibition. In contrast, under strong illumination, the elevated BG voltage drives the transistor into the accumulation regime, where the same TG pulse produces only weak inhibition. Consequently, brighter pixels retain their drain current, whereas dimmer pixels are progressively suppressed, thereby amplifying contrast between adjacent pixels under a fixed TG pulse. The transistor array thus processes the electrical signals generated by the photodiode array to produce a contrast-enhanced image. The bottom panels of Figure 1c show the photodiode array assembled on a PCB (left) and the fabricated adaptive inhibitory WSe₂ transistor array (right), confirming the hardware implementation.

2.2 | Operating Mechanism of Four-Terminal WSe₂ Transistor for Mode-Dependent Inhibition

In this adaptive inhibitory architecture, contrast amplification comes directly from the intrinsic transfer characteristics of the four-terminal WSe₂ transistor, without illumination-dependent bias adjustment. In contrast, conventional retinomorphic devices require deliberate gate-bias tuning for each illumination condition, rendering contrast enhancement a decision-dependent process. Our approach eliminates this requirement at the pixel level, enabling each pixel to autonomously adapt to its local illumination, making the approach scalable to array-level processing. In this section, we systematically analyze the physical operating principle of this transistor under controlled back-gate bias, revealing how its intrinsic transfer characteristics enable the autonomous inhibition response.

Figure 2a illustrates the four-terminal WSe₂ transistor, where the WSe₂ channel sits between top and bottom *h*-BN layers. The top *h*-BN is engineered as a charge-trapping layer via O₂ plasma treatment, while the BG controls the baseline drain current that sets the inhibition strength. The TG delivers V_{TG} pulses that trap electrons at the engineered defects, thereby suppressing the drain current and inducing inhibition. TEM/EDS reveals oxygen-related defects near the few-nm-thick WSe₂/top *h*-BN interface (Figure S6). Raman spectroscopy further confirms the defect formation through characteristic peak shifts in the *h*-BN E_{2g} mode after the O₂ plasma treatment (Figure S7) [48]. Direct electrical comparison between pristine and O₂-treated devices further reveals that inhibition emerges exclusively in the latter, confirming the trap-induced origin of the observed response (Figure S8).

To characterize the inhibition response, we apply identical V_{TG} pulses (5 V, 10 ms) at five V_{BG} levels of 3, 5, 8, 11, and 13 V (Figure 2b). Under low V_{BG} , the temporal drain current shows a pronounced reduction from $I_{D,pre}$ to $I_{D,post}$, whereas under high V_{BG} the same pulse induces only a marginal change. To quantify this behavior, we define the current inhibition ratio (CIR) as $I_{D,pre} / I_{D,post}$, which serves as a direct metric of the inhibition strength. As shown in Figure 2c, the CIR decreases monotonically from approximately 16.2 to 1.2 as V_{BG} increases from 3 to 13 V, demonstrating strong back-gate modulation of the inhibition strength. This monotonic trend suggests that the device transitions between distinct operating modes whose carrier-transport mechanisms respond differently to the same V_{TG} pulse.

To identify the physical origin of these two modes, we examine the transconductance (G_m) and the transfer characteristic (I_D - V_{BG}) at $V_D = 1$ V (Figure 2d). The conventional output and log-scale transfer characteristics of the WSe₂ transistor are further presented in Figure S9, showing linear output behavior and a clear off-state near $V_{BG} = 0$ V, respectively. Despite the intrinsic p-type tendency of WSe₂, the fabricated transistor exhibits N-type-dominant ambipolar behavior, which may be attributed to electron-beam-induced electron doping during the e-beam lithography process [49, 50]. At low V_{BG} (region i, blue), the transistor operates in the subthreshold mode, where carriers cross the Pt-WSe₂ Schottky barrier via thermionic emission, yielding

an exponential I_D - V_{BG} dependence (Figure 2e, top) [51, 52]. In contrast, at high V_{BG} (region ii, red), the channel becomes fully conductive, and charge transport is dominated by drift current, yielding a linear I_D - V_{BG} dependence (Figure 2e, bottom) [53, 54]. Within this framework, the V_{TG} pulse induces an effective gate voltage shift (ΔV) that further modulates I_D , equivalent to displacing the operating point from V_{BG} to $V_{BG} - \Delta V$. Analytical derivation in Note S3 yields $CIR \propto \exp(A\Delta V)$ for the subthreshold mode, with $A = q/nkT \gg 1$. For the accumulation mode, the same derivation gives $CIR \approx 1 + \Delta V/B$, where $B = V_{BG} - V_{TH} \gg 1$. Figure 2f visualizes these two regimes, showing that the same ΔV produces a much larger CIR in the subthreshold mode, with the difference further amplified by stronger V_{TG} stimulation. At small ΔV , the subthreshold slope scales as $A \gg 1$ while the accumulation slope scales as $1/B \ll 1$, so the two modes diverge already at the first pulse.

To verify this analytical prediction, we apply repeated V_{TG} pulses (5 V, 10 ms, 10 Hz) at two representative V_{BG} conditions (Figures 2g,h). At $V_{BG} = 3$ V (subthreshold mode), the drain current drops by nearly three orders of magnitude over pulse numbers (N) from 2 to 10, reflecting progressive trap filling (Figure 2g). In contrast, at $V_{BG} = 13$ V (accumulation mode), the same pulse train induces only a minor current variation, reflecting the reduced channel sensitivity in this regime (Figure 2h). Figure 2i summarizes the I_D evolution as a function of pulse number, where the mode-dependent inhibition difference becomes maximized at $N = 30$. At $V_{BG} = 8$ V, the response sits between these extremes, confirming a gradual transition between the two modes. Similar trends emerge when the V_{TG} pulse amplitude, width, or frequency is varied (Figures S10–S12), consistent with the larger ΔV each parameter induces. The same ΔV framework also indicates that the pulse duration is not an intrinsic limit of the inhibition: with a steeper transfer characteristic and a larger pulse amplitude, a single 250- μ s pulse is sufficient to induce mode-dependent inhibition, reducing the total operation time to approximately 1% of a typical camera frame interval (Figure S13 and Note S4) [55, 56].

Figure S14a further shows the time-dependent drain current measured for 100 s after the pulse stimulation under different V_{BG} conditions. The resulting inhibited state is retained over 100 s after the pulse train (Figure S14b), indicating non-volatile-like behavior that enables stable image processing. The transistor also maintained stable operation over 5,000 inhibition/recovery cycles (Figure S15), demonstrating the robust endurance and operational reliability of the proposed device. To examine whether this transport-regime-dependent inhibition is specific to the O₂-plasma-treated *h*-BN platform, we further fabricated an alternative *h*-BN/WSe₂/CIPS heterostructure in which the charge-trapping layer is replaced while the same WSe₂ channel is retained (Figure S16). The device exhibits pronounced counter-clockwise hysteresis, consistent with ferroelectric switching, and reproduces the distinct inhibition responses in the subthreshold and accumulation regimes under a fixed V_{TG} pulse condition. These results indicate that the inhibition originates from the transport regime of the channel rather than from a specific gate material. When V_{BG} is driven by a photodiode-generated voltage, this mode-controlled inhibition forms the basis of adaptive inhibition at the pixel level.

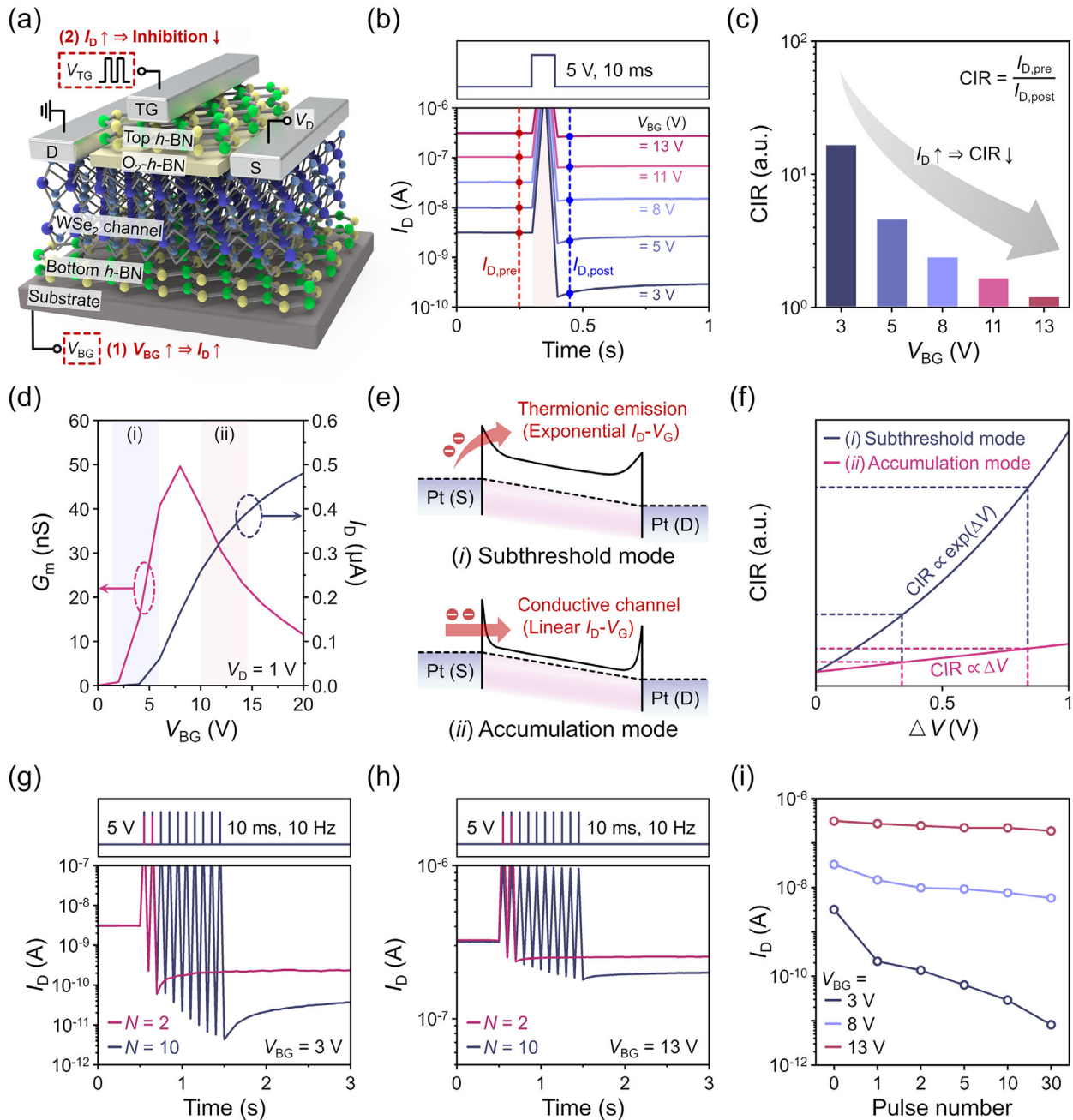


FIGURE 2 | Operating mechanism of four-terminal WSe₂ transistor for mode-dependent inhibition. (a) Schematic of the four-terminal device with top-gate (TG), source (S), drain (D), and back-gate (BG) terminals. (b) Temporal drain current response under identical V_{TG} pulses (5 V, 10 ms) at five V_{BG} values (3, 5, 8, 11, and 13 V). (c) Current inhibition ratio (CIR = $I_{D,pre} / I_{D,post}$) as a function of V_{BG} . (d) Transconductance (G_m) and transfer characteristic ($I_D - V_{BG}$) at $V_D = 1$ V, with regions (i) (blue) and (ii) (red) indicating the subthreshold and accumulation modes, respectively. (e) Band diagrams for the (i) subthreshold and (ii) accumulation modes. (f) Analytical CIR vs. effective gate-voltage shift (ΔV) for the two modes. (g, h) Drain current under repeated V_{TG} pulses (5 V, 10 ms, 10 Hz) at (g) $V_{BG} = 3$ V and (h) $V_{BG} = 13$ V. (i) I_D evolution with pulse number for $V_{BG} = 3, 8, \text{ and } 13$ V.

2.3 | Adaptive-Inhibition-Driven Contrast Enhancement and Noise Filtering Enabled by the Photodiode-Integrated Retinomorph In-Sensor Image Processor

Building on this mode-controlled inhibition, we now integrate the four-terminal WSe₂ transistor with a photodiode to realize adaptive inhibition at the pixel level. Figure 3a illustrates the overall operating concept of the proposed retinomorph in-

sensor image processor. Here, a blurred input image, illustrated by a noisy “G” pattern, is first projected onto a 5×5 photodiode array. Each photodiode converts its local light intensity into a back-gate voltage (V_{BG}) that drives the corresponding adaptive inhibitory transistor in the underlying 5×5 device array. A common V_{TG} pulse train applied across the array then induces adaptive inhibition at each pixel, whose strength scales with the local V_{BG} . Noise pixels with weak optical inputs yield low V_{BG} , holding the transistor in the subthreshold mode where the

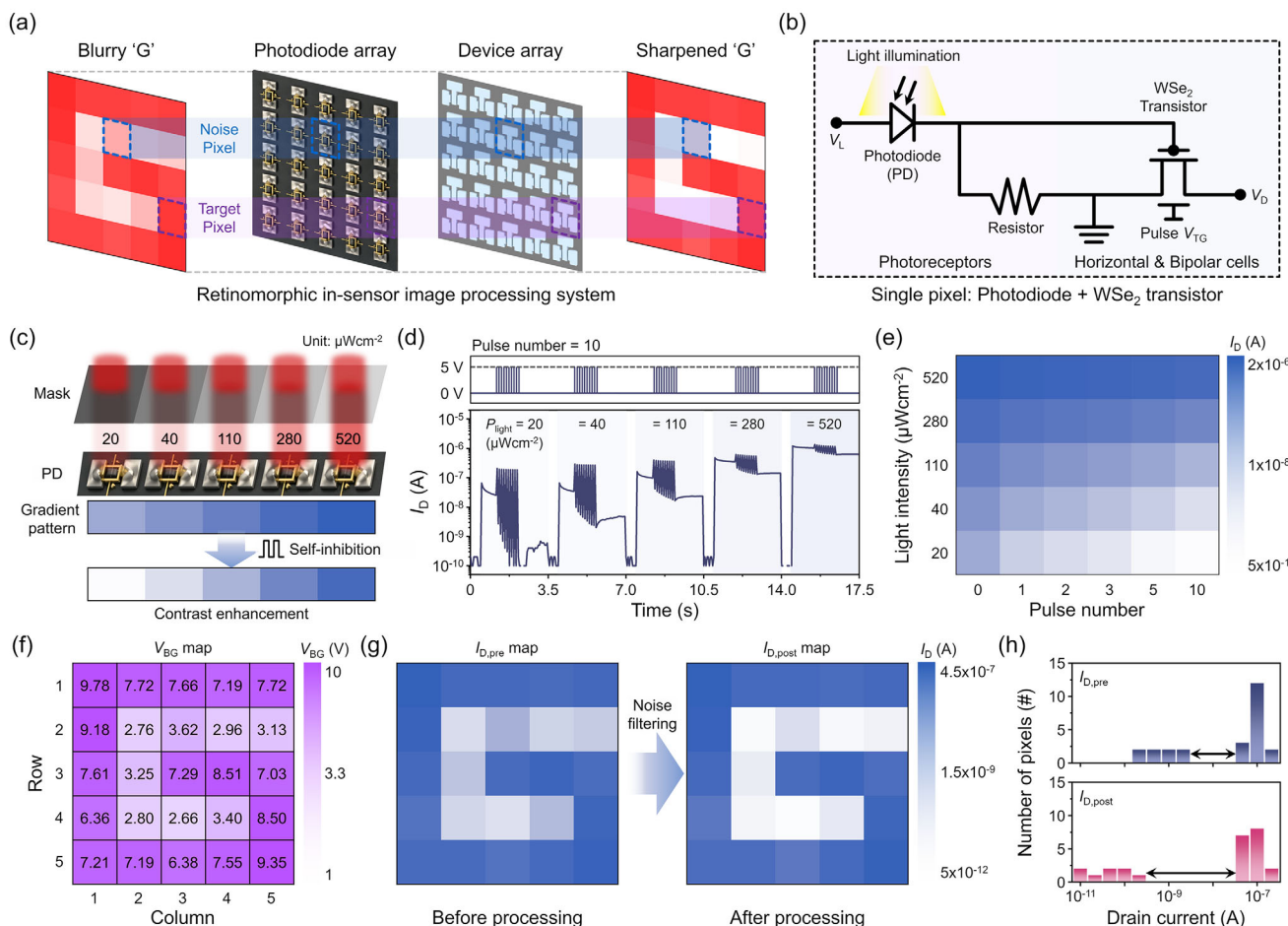


FIGURE 3 | Adaptive-inhibition-driven contrast enhancement and noise filtering enabled by the photodiode-integrated retinomorphic in-sensor image processor. (a) Schematic of the proposed in-sensor image processing system, where a noisy “G” input is sharpened by cascaded 5×5 photodiode and adaptive inhibitory transistor arrays. (b) Single-pixel circuit comprising a photodiode (PD), a load resistor, and a four-terminal WSe₂ transistor ($V_L = 20$ V, $V_D = 1$ V, $R = 1$ M Ω). (c), Schematic illustration of adaptive-inhibition-driven contrast enhancement, where a transmittance-graded mask generates five light intensities (20, 40, 110, 280, and 520 $\mu\text{W cm}^{-2}$) incident on a single row of photodiodes. (d) Drain current response across different light intensity conditions in the gradient-pattern under repeated V_{TG} pulse stimulation (5 V, 10 ms, 10 Hz). (e), Drain current mapping as a function of pulse number and light intensity. (f), V_{BG} map generated from a noisy “G” pattern illumination on a 5×5 photodiode array. (g), Drain current maps before ($I_{D,pre}$) and after ($I_{D,post}$) image processing for the noisy “G” pattern. (h) Histogram of drain current distributions before and after processing. The current range is divided into three logarithmic bins per decade (1–2.16, 2.16–4.67, and 4.67–10).

V_{TG} pulse strongly suppresses its drain current. Target pixels, however, receive stronger illumination that elevates V_{BG} into the accumulation mode, where the drain current is largely preserved under the same pulse. Through this spatially adaptive inhibition, the system selectively suppresses noise while preserving target features, producing the sharpened “G” output image. The single-pixel circuit underlying this operation is shown in Figure 3b.

At the single-pixel level, a commercial Si photodiode (TEMD7000×01, Vishay Semiconductors) and a load resistor ($R = 1$ M Ω) are connected in series between load voltage ($V_L = 20$ V) and ground. The midpoint of this divider sets V_{BG} of the adaptive inhibitory transistor, which tracks the photodiode resistance and hence the incident light intensity. Under dark conditions, the photodiode resistance is high, so V_L drops mostly across the photodiode and V_{BG} remains near zero, keeping the transistor OFF. Upon illumination, the photodiode resistance decreases and the voltage drop redistributes toward R , raising

V_{BG} linearly with light intensity over two decades from 10 to 520 $\mu\text{W cm}^{-2}$ (see Figure S17). This topology mirrors the early visual pathway, with the photodiode acting as the photoreceptor and the resistor-transistor pair implementing horizontal and bipolar cell functions. Stable V_L sweep operation is further confirmed at 600 $\mu\text{W cm}^{-2}$ (see Figure S18), providing a robust pixel-level interface for the array-level demonstrations that follow.

Building on this pixel-level interface, we integrated the 5×5 photodiode array with a single adaptive inhibitory transistor through multiplexed control, addressing each pixel sequentially with low inter-pixel crosstalk. The 25 photodiodes were connected to the same transistor one at a time, and the 25 processed outputs were mapped back to their spatial positions to reconstruct the processed image. The control architecture, with column-row multiplexers driven by a Bluetooth-enabled SoC, is detailed in Figure S19 and Note S5. The fidelity of this addressing is verified in Figure S20, where in-pixel current dominates over transfer to

neighboring pixels with negligible crosstalk. Figure 3c illustrates the gradient experiment, where a transmittance-graded mask delivers five intensities of 20, 40, 110, 280, and 520 $\mu\text{W cm}^{-2}$ to one row of photodiodes. Figure S21 details the mask alignment and optical configuration, and these intensities translate into V_{BG} values from 1 to 20 V spanning both the subthreshold and accumulation modes. Before any V_{TG} pulse, the five pixels already produce distinct drain currents that follow their respective V_{BG} with modest inter-pixel spread. V_{TG} pulses of 5 V, 10 ms, and 10 Hz then trigger mode-dependent adaptive inhibition, with Figure 3d showing subthreshold pixels strongly suppressed while accumulation-mode pixels remain largely unchanged. Figure 3e summarizes this trend with an I_{D} map across pulse number and intensity, complemented with time-domain traces at $N = 1, 2, 3$, and 5 in Figure S22. After 10 pulses, the inter-pixel current spread expands to nearly five orders of magnitude, confirming adaptive-inhibition-driven contrast enhancement of the gradient pattern.

We then extend this validation from gradient illumination to a full noisy “G” pattern projected across the entire 5×5 array. Figure 3f shows the resulting V_{BG} distribution, where pixels along the “G” contour exceed 6 V (peaking at ~ 9.8 V) while background and noise pixels remain near 2 to 3 V. Applying the same V_{TG} pulse train then produces the $I_{\text{D, pre}}$ and $I_{\text{D, post}}$ maps in Figure 3g, where noise pixels are selectively suppressed while target pixels remain intact, sharpening the “G” contour. Figure S23 quantifies this behavior, with target pixels retaining I_{D} near 10^{-7} A and noise pixels dropping by one to two orders of magnitude after processing. This selective suppression increases the target-to-noise contrast ratio to approximately two orders of magnitude. The resulting bimodal pixel distribution is shown in Figure 3h. Figure S23c also reveals a monotonic CIR decrease with increasing light intensity, reproducing the mode-dependent trend established at the device level in Figure 2c. The system also exhibits recovery behavior after the inhibition process. The inhibited current can be restored to its initial level by applying a single, longer negative recovery pulse under different illumination conditions (Figure S24). Furthermore, the inhibition–recovery responses show good overlap over 10 cycles at each illumination condition, with the illumination-dependent inhibition ratio preserved throughout, confirming the reproducibility of the adaptive inhibition–recovery operation. These results validate the integrated system as a functional retinomorphic in-sensor image processor, with array-scale reproducibility further examined below.

The device reproducibility was further evaluated at the array level using a 3×2 WSe₂ transistor array. The fabrication processes are detailed in Figures S25 and S26 and in the Method section. The six transistors exhibited consistent output and transfer characteristics, demonstrating device-to-device uniformity across the array (Figure S27). Furthermore, under the same V_{TG} pulse train, all devices consistently exhibited stronger inhibition in the subthreshold regime and weaker inhibition in the accumulation regime (Figure S28). These results confirm the device-to-device reproducibility of the transport-regime-dependent inhibition behavior. When integrated with a photodiode array, this 3×2 transistor array reproduces adaptive-inhibition-driven contrast enhancement on a “V” pattern, as shown in Figure S29. These results support the reproducibility of the proposed retinomorphic in-sensor image processor at the array level, paving the way

for its application to low-visibility imaging tasks. The present system is assembled from mechanically exfoliated WSe₂ flakes together with discrete photodiode and resistor components, which restricts the fabricated transistor array to 3×2 . Enlarging the array introduces inter-pixel crosstalk, device-to-device yield over larger areas, and an increased footprint of the peripheral selection circuitry, none of which are addressed by the present data. Monolithic integration based on wafer-scale growth of the 2D channel, together with pixel-level integration of the photodiode and load elements, will therefore be required for larger arrays. To further improve scalability toward large-scale integration, we propose a compact pixel architecture in which the discrete components are replaced with CVD-grown 2D devices and integrated within a single pixel, as further discussed in Figure S30 and Note S6.

2.4 | Adaptive-Inhibition-Based Image Processing for ANN-Based License-Plate Recognition Under Low-Visibility Conditions

Next, we numerically simulate the application of adaptive-inhibition-based processing to license-plate recognition under adverse weather conditions. Figure 4a depicts a vehicle license plate degraded under rain and fog, with the magnified Noisy Patch inset highlighting scattering-induced noise at the single-character level. Reliable license-plate recognition under such conditions is a long-standing challenge for outdoor edge-vision tasks in autonomous driving and traffic monitoring, where weather-induced degradation directly compromises recognition accuracy [57]. In the application considered here, the recognition system is trained under normal conditions and deployed with fixed weights; when rain or fog obscures the plate, the same classifier can no longer resolve the characters. The proposed in-sensor processing restores the degraded patches before they reach the classifier, so that recognition is recovered without retraining or modifying the network. Figure 4b illustrates the processing and classification framework, in which 7×7 noisy patches are extracted from the degraded plate, processed by the simulated retinomorphic array, and classified by an artificial neural network (ANN). During processing, low-intensity background pixels are selectively suppressed while high-intensity character pixels are preserved, producing the restored patches that serve as the ANN input. The retinomorphic processing uses experimentally calibrated inhibition characteristics of the four-terminal WSe₂ transistor, ensuring the simulation reflects the measured device response. The dataset comprises ten classes (“1”, “2”, “3”, “7”, “A”, “B”, “C”, “H”, “S”, and “T”). The 10,000 training patches were clean, noise-free character images, whereas the 2,000 test patches were generated from the same clean characters by applying a synthetic Gaussian noise model. Classification is performed by a multilayer perceptron (MLP) with 49 input neurons, 16 hidden neurons, and 10 output neurons, with the training and evaluation protocol detailed in Methods.

Figure 4c illustrates the pulse-by-pulse processing of a representative “B” patch, beginning with the heavily noisy input in which the character is barely distinguishable from the background. After the first V_{TG} pulse, the low-intensity background pixels are partially suppressed while the high-intensity character pixels remain, revealing the initial “B” outline. By the tenth pulse,

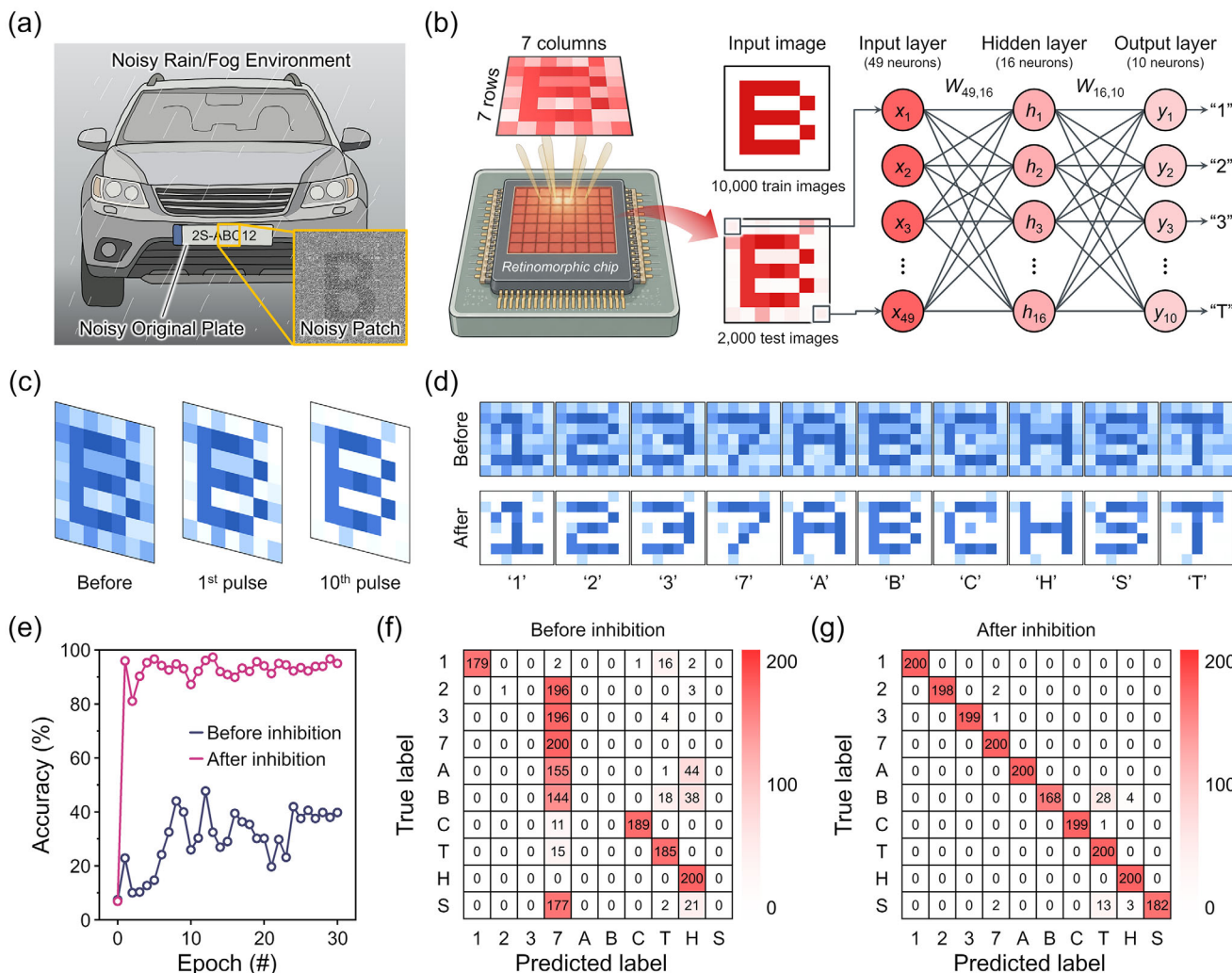


FIGURE 4 | Adaptive-inhibition-based image processing for ANN-based license-plate recognition under low-visibility conditions. (a) Example of blurred vehicle license plates captured under adverse weather conditions. (b) Schematic illustration of the retinomorphic in-sensor image processor and multilayer perceptron, showing the offline training stage on clean characters and the subsequent inference stage in which the fixed network classifies the processed inputs. (c) Evolution of the noisy "B" character patch before processing, after the first pulse, and after the 10th pulse application. (d) Comparison of character patches from ten different classes before and after image processing. e, Classification accuracy as a function of training epoch for the network trained on clean characters, evaluated on the test sets before and after image processing. (f, g) Confusion matrices for ten different character classes before and after image processing, respectively.

the "B" pattern is fully resolved, with the noisy background nearly eliminated and the character boundaries sharply defined. Figure 4d extends this comparison to the full ten-class dataset, with each unprocessed patch heavily obscured by noise and each processed patch recovering the distinguishing morphological features of its character. The consistent recovery across all ten classes confirms that adaptive inhibition processes noise patterns independently of character morphology, showing that the approach generalizes across character classes.

Figure 4e shows the classification accuracy vs. training epoch for the network trained on clean characters, evaluated on the noisy test set before processing and on the test sets after processing. Accuracy on the unprocessed test set saturates at only 47.7% because of the limited contrast, whereas accuracy on the test set processed by the tenth pulse reaches 97.3%. Figure S31 traces this 49.6 percentage-point gain pulse by pulse, showing monotonic

improvement with the first pulse alone lifting accuracy by ~25.1 percentage points. Before processing, Figure 4f reveals frequent misclassification of similar-morphology characters (notably 2, 3, A, B, and S) into the "7" class, indicating noise-induced collapse of distinguishing features. After processing, Figure 4g shows that most predictions concentrate along the diagonal, indicating that the enhanced contrast restores the morphological features required for class separation. Figure S32 traces this transition pulse by pulse, with the confusion matrix progressively approaching an ideal diagonal form as inhibition strengthens. These results show that adaptive-inhibition-based processing can restore noise-corrupted images and improve downstream classification. To examine whether this behavior extends beyond the 7 × 7 license-plate patches, we applied the same processing to the MNIST 28 × 28 grayscale dataset, where the classification accuracy increased from 16.47% to 96.19% after ten V_{TG} pulses (Figures S33 and S34).

To further benchmark the proposed system against previously reported retinomorphic devices, we compared key metrics including device area, processing latency, energy consumption, illumination range, current modulation ratio, recognition performance, and decision-circuit requirements (Table S1). The optimized operation reduces the processing latency to 0.25 ms while achieving a current modulation ratio of ~ 149 and improving the simulated MNIST recognition accuracy from 33.07% to 95.71% with a single pulse (Figures S13 and S35). Although the present implementation incorporates a photodiode and load resistor, the proposed architecture enables intensity-adaptive processing directly through an analog voltage-divider without intermediate decision-making or pulse-selection circuitry. In contrast, decision-based approaches require additional circuit blocks even for binary selection, resulting in increased hardware and energy overhead, which can further increase for more complex decision functions, such as selecting different gate voltages according to the illumination level [5, 43, 58]. These characteristics highlight the potential of the proposed system for fast, energy-efficient, and decision-free retinomorphic in-sensor image processing.

3 | Conclusion

In summary, we demonstrated an adaptive inhibitory retinomorphic in-sensor image processor capable of adaptive and decision-free in-sensor image processing. Inspired by the contrast-enhancement mechanism of the biological retina, the proposed system integrates a photodiode with a four-terminal WSe₂ transistor in a tightly coupled cascade for sensing and inhibition. Unlike conventional retinomorphic systems that require illumination-dependent gate-bias tuning, our system autonomously modulates its inhibition strength in response to illumination conditions. Electrical analysis revealed that the adaptive inhibition originates from the intrinsic transfer characteristics of the transistor, which transitions between subthreshold and accumulation regimes depending on the back-gate voltage. Specifically, as the back-gate voltage increased from 3 to 13 V, the current inhibition ratio decreased monotonically from ~ 16.2 to ~ 1.2 , demonstrating strong back-gate modulation of the inhibition strength. Consequently, weakly illuminated pixels undergo strong inhibition, whereas brightly illuminated pixels preserve their current levels, enabling selective suppression of background noise while maintaining target features. Leveraging this principle, the 5×5 integrated array achieved contrast enhancement on a gradient pattern and noise filtering on a noisy “G” input. Furthermore, the fabricated 3×2 adaptive inhibitory transistor array exhibited consistent transfer characteristics and inhibition responses, confirming device-to-device uniformity at the array level. Finally, simulations of license-plate recognition under low-visibility conditions showed that adaptive-inhibition-based processing restores degraded character morphology, improving ANN classification accuracy from 47.7% to 97.3%. These results show that adaptive inhibition offers a practical and scalable route to adaptive in-sensor image processing, making it a strong candidate for edge-vision hardware operating in low-visibility environments. It is worth noting that the proposed system emulates only one aspect of retinal processing. The excitatory responses of the retina and the center-surround processing that arises from interactions among photoreceptors and downstream cells are not reproduced, and such spatial operations are left to the downstream processor.

Extending the present in-pixel operation toward these spatial functions, for example by coupling neighboring pixels at the array level, remains a direction for future work.

4 | Experimental Methods

4.1 | Fabrication of the Adaptive Inhibitory Device

The *h*-BN and WSe₂ flakes (2D semiconductors) were mechanically exfoliated and transferred onto a 90-nm-thick silicon dioxide (SiO₂)/heavily p-doped Si substrate. The source and drain (S/D) electrode regions, defined within a channel length of ~ 10 μm and a channel width of ~ 5 μm , were patterned on polymethyl methacrylate resist (950 PMMA A6) using electron-beam lithography (EBL; Raith Elphy Quantum graphic writer) performed with a field-emission scanning electron microscope (SU5000). Subsequently, 10/100-nm-thick platinum/silver (Pt/Ag) electrodes were deposited via e-beam evaporation (Solar-bridge). The top *h*-BN flake was first transferred onto a polydimethylsiloxane (PDMS) film and subsequently treated with O₂ plasma using a reactive ion etching (RIE) system (PLASMA LAB80) under conditions of 50 sccm gas flow rate, 10 W RF power, and 30 s etching duration to form the charge-trapping layer. After plasma treatment, the top *h*-BN was transferred onto the device with pre-fabricated S/D electrodes. The top-gate electrode, defined with a width of ~ 5 μm , was patterned on 950 PMMA A6 using the same EBL process. Finally, 30/300-nm-thick Pt/Ag electrodes were deposited via e-beam evaporation to form the top-gate electrode.

4.2 | Electrical Characterization of the Adaptive Inhibitory Transistor

The current-voltage ($I_{\text{DS}}-V_{\text{GS}}$) characteristics and retinomorphic responses of the device were characterized using a parameter analyzer (Keysight B2912B) together with a waveform generator (Keysight 33500B) in a vacuum probe station ($\sim 1 \times 10^{-5}$ Torr). For processing operation, electrical spike stimuli were applied to the top-gate terminal in the form of square-wave pulses with controlled amplitude, frequency, width, and pulse number.

4.3 | Material Characterization of *h*-BN/WSe₂/*h*-BN Heterostructure

For the structural and elemental analyses of the *h*-BN/WSe₂/*h*-BN heterostructure, TEM lamellae were prepared via focused ion beam (FIB) milling using a Hitachi NX5000 system. The prepared lamellae were subsequently characterized using a Tecnai G2 F30 S-Twin TEM operated at an accelerating voltage of 300 kV. Raman spectra were obtained using an XperAM C system (Nanobase) equipped with a 532 nm excitation laser.

4.4 | Fabrication and Measurement of the Photodiode-Integrated Retinomorphic in-Sensor Image Processor

The photodiode array readout system was implemented on an FR-4-based printed circuit board (PCB) with dimensions of

$60 \times 40 \times 1.2 \text{ mm}^3$, designed using Fusion. External power was supplied using a benchtop power supply (WANPTEK). To ensure stable operation of the low-voltage circuitry, a low-dropout regulator (TPS7A033PDQNR, Texas Instruments) was employed to convert the input voltage into a stable 3.3 V DC supply for the overall system. In addition, a boost converter (TPS61041DRVR, Texas Instruments) was used to generate the 20 V bias voltage required for photodiode operation. Row and column selection within the photodiode array was controlled by a Bluetooth Low Energy (BLE) System-on-Chip (SoC; ISP1807, Insight SIP). Digital select signals generated by the BLE SoC were applied to two 8:1 multiplexers (MUXs; CD4051BQWR, Texas Instruments), where five input channels from each MUX were utilized to address the 5×5 photodiode array. Through the MUX-based row and column selection scheme, one terminal of the selected photodiode cell was connected to the 20 V bias line, while the opposite terminal was connected to a reference resistor (PDB12-M4251-105BF, Bourns Inc.) to form a voltage divider that generated the photodiode-dependent V_{BG} signal. The firmware for the BLE SoC was programmed using Segger Embedded Studio, and a custom iOS-based user interface was developed using Xcode, enabling real-time wireless control of the MUX select signals through BLE communication. For retinomorphic characterization, the system operating parameters were configured at $V_L = 20 \text{ V}$, $V_D = 1 \text{ V}$, and $R = 1 \text{ M}\Omega$. A 625 nm LED (M625L4, Thorlabs), modulated by an LED driver (LEDD1B, Thorlabs), served as the optical input source. The optical illumination was applied in a free-space, normal-incidence configuration to uniformly cover the entire photodiode array. To evaluate image processing performance, 5×5 pixel patterns (representing the letter “G” or gradient profiles) were printed on overhead projector (OHP) films to serve as optical masks, which were placed directly atop the photodiode array. The transient retinomorphic responses of the system were characterized using the parameter analyzer together with the waveform generator in the vacuum probe station.

4.5 | Fabrication of the Adaptive Inhibitory Device Array

First, contact pads (Ti/Au, 4/40 nm) and $\sim 5\text{-}\mu\text{m}$ -wide bottom-gate electrodes (Ti/Au, 4/40 nm) were sequentially fabricated on a 90-nm-thick SiO_2 /heavily p-doped Si substrate using EBL and e-beam evaporation (Steps 1 and 2). Subsequently, large-scale *h*-BN and WSe_2 flakes were mechanically exfoliated and sequentially transferred onto the pre-patterned bottom-gate array using a PDMS stamp (Steps 3 and 4). To define individual devices within the 3×2 configuration, the transferred *h*-BN/ WSe_2 heterostructure was patterned by EBL into six distinct channels with a length of $\sim 18 \text{ }\mu\text{m}$ and a width of $\sim 4 \text{ }\mu\text{m}$ (Step 5). Inductively coupled plasma reactive ion etching (SPIRAL 6C) with SF_6 plasma was then performed to remove the unmasked 2D regions, ensuring electrical isolation between neighboring pixels (Step 6). Following channel isolation, common source and individual drain electrodes were defined. The S/D electrodes (Pt/Ag, 10/100 nm) were patterned via EBL and e-beam evaporation, yielding a channel length of $\sim 11 \text{ }\mu\text{m}$ (Step 7). To incorporate the charge-trapping layer, an additional large-scale *h*-BN flake was exfoliated onto a PDMS film and subjected to O_2 plasma treatment via RIE (50 sccm gas flow rate, 10 W RF power, and 30 s etching duration). The plasma-treated *h*-BN layer was then transferred over the entire $3 \times$

2 device area (Step 8). Finally, common top-gate electrodes ($\sim 6 \text{ }\mu\text{m}$ wide) were patterned via EBL, followed by e-beam evaporation of a 30/300-nm-thick Pt/Ag stack, completing the fully integrated 3×2 adaptive inhibitory transistor array (Step 9).

4.6 | Retinomorphic Simulation for Vehicle License Plate Recognition Under Adverse Weather Conditions

To evaluate vehicle license plate recognition performance in noisy environments, MLP simulations were performed based on the adaptive inhibition of the retinomorphic in-sensor image processor. The dataset consisted of 10,000 training images and 2,000 test images across 10-character classes commonly used in vehicle license plates, where each image was formatted to a resolution of 7×7 pixels (49 pixels total). The initial pixel intensities (0–255) were quantized into eight discrete levels and mapped to experimentally measured current levels of the adaptive inhibitory transistor (0.1, 0.3, 1, 3, 10, 30, 100, and 300 nA). The inhibition behavior corresponding to each current level was experimentally characterized in advance and assigned to individual pixels according to their mapped current levels. The resulting inhibited current states were then converted into normalized pixel intensities within the range of [0, 1] through log-scale normalization, thereby reconstructing the processed noise-mitigated images. The neural network was implemented as an MLP receiving 49-pixel patches as inputs. The MLP consisted of an input layer with 49 neurons, a hidden layer with 16 neurons, and an output layer with 10 neurons. ReLU and Softmax functions were employed as the activation functions for the hidden and output layers, respectively. The MLP was trained on the 10,000 clean, noise-free character images. After training, the network weights were fixed, and no retraining or fine-tuning was performed at any stage. The same fixed network was then evaluated on the 2,000 noise-degraded test images and on the corresponding test sets obtained after applying one to ten V_{TG} pulses in the simulated in-sensor processing, so that the classifier remained identical across all conditions.

4.7 | Simulation for MNIST Handwritten-Digit Recognition

For the MNIST simulation, a fully connected multilayer perceptron was employed. Each 28×28 -pixel image was flattened into a 784-D input vector and fed into a hidden layer containing 128 neurons with a rectified linear unit (ReLU) activation function. The output layer consisted of 10 neurons corresponding to the 10-digit classes, with a Softmax activation function. The standard MNIST dataset was divided into 60,000 training images and 10,000 test images. The network was trained by minimizing the categorical cross-entropy loss using the Adam optimizer with a learning rate of 2×10^{-3} and a batch size of 128.

Author Contributions

J.B., J.O., S.O., and D.-H.K. conceived the work. J.B., J.O., S.O., and D.-H.K. designed the experiments and analyzed the data. J.O. and S.O. implemented 5 by 5 retinomorphic hardware. J.B., M.P., H.C., and T.K. fabricated adaptive inhibitory WSe_2 device and array. J.O., H.K., and

S.O. designed neural network simulation framework and conducted the simulations. J.Y.P. and H.-J.S. conducted Raman spectroscopy. J.B. and H.S. analyzed TEM. J.B., J.O., Y.Z., S.O., and D.-H.K. prepared the manuscript. All authors revised the manuscript. S.O. and D.-H.K. guided the entire project.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adfm78443-sup-0001-SuppMat.docx.